Pulse Guard AI

Intelligent Observability & Event Watchdog

Graduate Vibe Coding Challenge — Project 3 (SRE)

Architect-directed. AI-engineered. Human-in-the-loop.

The Problem

Modern cloud platforms emit **millions of log lines**.
Buried inside are the
error spikes that precede outages.

- 🔍 Manual log-watching doesn't scale
- ⏰ Static thresholds are noisy & miss creeping degradation
- 🚨 Alerts arrive **after** customers already feel the pain

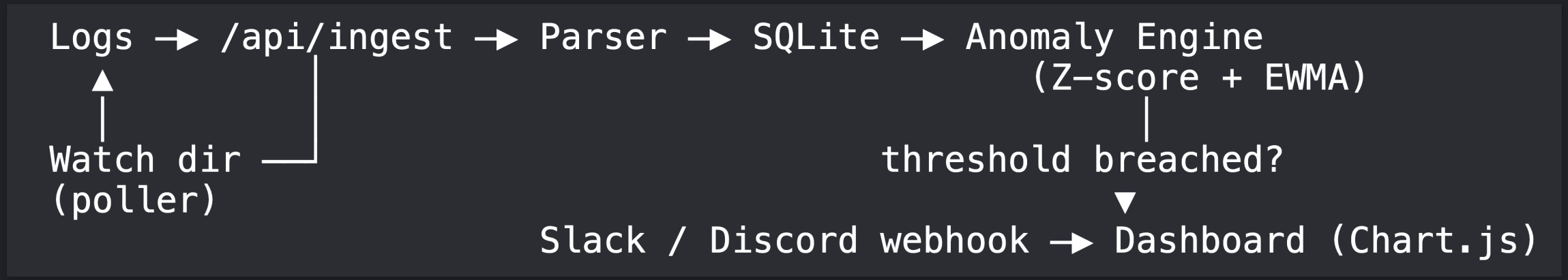
> SREs need an **automated watchdog** that learns normal and flags abnormal.

The Vision

An **API-first**, **AI-driven** service that:

1. **Ingests** application/platform logs (any format)
2. **Learns** each service's normal error baseline
3. **Detects** anomalies / spikes with explainable statistics
4. **Alerts** via real Slack / Discord webhooks
5. **Visualizes** health trends on a live dashboard

Architecture



API-first · Free DB · Explainable AI · Real alerts

Tech Stack

Layer	Technology	Why
API	FastAPI + Uvicorn	Async, auto OpenAPI docs
Database	SQLite + SQLAlchemy 2	Free-tier, zero-config
AI Logic	Rolling Z-score + EWMA	Explainable, dependency-light
Scheduler	APScheduler	Continuous log polling

AI Logic — How Detection Works

For each service, per 60-second bucket of **error** events:

- **EWMA baseline** adapts to traffic: `ewma = α·count + (1−α)·ewma`
- **Rolling Z-score**: `z = (count − mean) / std`
- **Spike** flagged when `count ≥ 5` **and** `z ≥ 3` **and** `count > baseline`
- **Severity**: `z ≥ 6` → 🔴 critical, `z ≥ 3` → 🟠 warning

> Every alert carries a **human-readable reason** — no black box.

Anomaly Coverage (Enterprise Test Data)

10-service, ~13k-line synthetic dataset — **8 fault patterns + 2 healthy**:

Pattern	Example	Result
Sudden spike	payment-svc	🚨
Gradual creep	search-svc	🚨
Sustained outage	db-proxy	🚨
Cascade	gateway → auth	🚨

Continuous Watchdog Mode

Background **APScheduler** tails `*.log` files in a watch directory:

- ⚡ Ingests only **new** bytes each tick (offset tracking)
- 🔄 Handles **log rotation** / truncation
- 🧠 Runs detection + fires alerts automatically
- 🎛️ Controllable via API & dashboard (start / stop / poll now)

```
PULSE_POLL_ENABLED=true uvicorn app.main:app
echo "... ERROR [svc] boom" >> ingest_watch/svc.log   # auto-detected
```

Real Alerting — Slack & Discord

Auto-detects the webhook flavour from its URL:

- **Slack** → Block Kit incident message
- **Discord** → color-coded rich embed
- **Generic** → raw incident JSON
- **No URL** → local sink (fully demoable offline)

```
PULSE_WEBHOOK_URL=https://hooks.slack.com/services/T/B/X \
  uvicorn app.main:app
```

Live Dashboard

- 📊 **Health score** + error-rate trend chart (Chart.js)
- 🚨 Detected anomalies table (severity, z-score, baseline)
- 📨 Fired webhook alerts log
- 🎛️ Poller controls + one-click demo / enterprise data
- 🔄 Auto-refresh every 15s

Quality & Testing

- ✅ **22 automated tests** (pytest) — parser, detection, scenarios,
  scheduler tailing/rotation, Slack/Discord payloads
- ✅ True-positive **and** true-negative anomaly coverage
- ✅ Isolated test DB, deterministic synthetic data
- ✅ Graceful degradation (webhook failures never crash ingestion)

Vibe Coding Workflow

- 🧭 **Architect** sets vision & rules; **AI** writes 100% of code
- 📝 `prompts.md` — full human-in-the-loop audit log
- ⏱️ MVP delivered well within the 4–6h target window
- 🔁 Bugs fixed by describing them to the AI — **zero manual edits**

Roadmap

- 🤖 ML detectors (seasonal Holt-Winters, isolation forest)
- 🔗 Native cloud log sources (CloudWatch, Azure Monitor)
- 📈 Per-metric detection (latency, throughput, saturation)
- 🧵 Incident correlation across cascading services
- 🐳 Container + Postgres deployment

Pulse Guard AI

Detect the spike. Trigger the alert. Protect the pulse.

Thank you.

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